PROJECT REPORT

A Self-Reflective Large-Language-Model Agent for Explainable Customer Purchase Intention Prediction

A reproducible, explainable, and authenticated machine-learning workspace for customer decision support

Project type	Applied AI / Machine Learning prototype
Core approach	Behavioural prediction + explanations + self-reflection
API framework	FastAPI
User interface	Responsive browser workspace
Programming language	Python
Deployment	Vercel production environment

Machine Learning • Explainable AI • Human-supervised Decisions

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EXECUTIVE SUMMARY

This project implements an end-to-end agent for predicting customer purchase intention from behavioural session features. It combines a calibrated logistic-regression model with signed evidence, counterfactual guidance, and a deterministic self-reflection pass. The result is a decision-support system that exposes not only what it predicts, but also why the prediction was made and whether a human should review it.

The project is intentionally local-first and open source. It runs without a paid API or proprietary service, provides a reproducible synthetic dataset for experimentation, and deploys a lightweight inference adapter to Vercel. The hosted workspace uses credential-based access and returns exactly three outputs for each scored session: intention, evidence, and reflection.

The current verified model run achieved a ROC-AUC of 0.7949, accuracy of 0.7375, and Brier score of 0.1801 on 160 held-out samples. These results validate the pipeline; they do not replace evaluation on consented production data.

01 PROJECT OVERVIEW

Problem statement, objectives, and users

Problem Statement

Customer purchase-intention systems often reduce a complex behavioural session to a single score. A score without evidence is difficult to audit, difficult to improve, and easy to over-trust. This project addresses the need for a transparent prediction workflow that gives decision makers measurable evidence and an explicit confidence check.

Project Objectives

Objective	Outcome
Predict	Estimate the likelihood of purchase from session behaviour.
Explain	Expose signed factors that increase or decrease the score.
Reflect	Identify uncertainty, invalid inputs, and boundary cases.
Protect	Require credentials before metrics and predictions are accessible.
Deploy	Provide a free, reproducible production URL and local run path.

Intended Users

The project is designed for researchers, customer-experience analysts, and engineers who need an inspectable baseline before introducing a more complex model. It is not intended to make irreversible decisions about individuals or to replace human review.

Design Principle

Every prediction should answer three questions: What is the intention estimate? What behavioural evidence supports it? How certain is the system, and what should happen next?

02 SYSTEM DESIGN

From behavioural input to a deployable workspace

Architecture

The local research path uses Python, pandas, NumPy, scikit-learn, FastAPI, SQLite, and vanilla HTML/CSS/JavaScript. The hosted path uses FastAPI, Pydantic, and standard-library Python to remain below Vercel's serverless function-size limit. Both paths preserve the same three-output contract.

Layer	Responsibility	Project surface
Data	Generate and validate privacy-safe aggregates	data.py
Model	Train and persist classifier	model.py
Explain	Calculate signed feature contributions	explain.py
Reflect	Check confidence and stability	reflection.py
Serve	Expose API, authentication, and audit store	api.py / api/index.py
Interface	Collect session features and render results	static/index.html

End-to-End Flow

1. A user signs into the workspace. 2. Session features are validated. 3. The model calculates a purchase probability. 4. Feature contributions are ranked by impact. 5. Reflection checks confidence and data quality. 6. The API returns intention, evidence, and reflection. 7. A local audit record is written to SQLite or ephemeral Vercel storage.

Data Features

The feature set includes session duration, pages viewed, product views, cart additions, prior purchases, discount usage, device type, traffic source, and returning-customer status. Direct personal identifiers are not part of the demonstration schema.

03 MODEL AND EXPLAINABILITY

A measurable baseline with visible reasoning

Data Generation

When private data is unavailable, the generator produces a reproducible behavioural dataset using a fixed random seed. The target is a purchase observed within the chosen session window. This supports pipeline development while making the limitations of synthetic evaluation explicit.

Prediction Model

The local model is a logistic-regression pipeline. Numeric inputs are standardized and categorical values are one-hot encoded. For an input vector x, the probability is $p(y=1|x) = 1 / (1 + \exp(-(w^T x + b)))$. The probability is converted into low, medium, or high risk bands for the interface.

Three Outputs

Output	Returned content
1. Intention	Purchase probability and risk band.
2. Evidence	Positive and negative behavioural factors.
3. Reflection	Confidence, review flag, and next action.

Self-Reflection

The reflection layer checks input quality, confidence away from the 0.5 boundary, explanation availability, and boundary stability. A low-confidence case is not presented as a definitive conclusion. Instead, it is marked for human review and accompanied by a practical counterfactual suggestion.

Explainability Boundary

Feature contributions explain the model's internal score; they do not prove that changing a feature will cause a purchase. The report therefore describes counterfactuals as intervention hypotheses, not causal findings.

04 SECURITY AND DEPLOYMENT

Credential protection and production delivery

Authentication

The hosted dashboard uses environment-backed APP_USERNAME, APP_PASSWORD, and SESSION_SECRET values. Successful login creates an HTTP-only, signed cookie with an eight-hour lifetime. The /predict and /model/metrics routes return 401 without a valid session. Health and login remain public so service availability can be checked.

Control	Implementation
Credentials	Read from Vercel environment variables; never committed.
Session	HMAC-signed HTTP-only cookie.
Transport	Secure cookie flag on Vercel HTTPS.
Data	Behavioural aggregates; no direct identifiers.
Audit	SQLite locally; ephemeral /tmp on Vercel.
Production gap	Use durable access-controlled database for real use.

Deployment

Production URL: https://purchase-intention-agent.vercel.app. The repository includes Vercel configuration, a lightweight Python entrypoint, and a small requirements file. Local research and training continue to use the full scientific Python stack.

Operational Note

Vercel function storage is ephemeral. A production system should add durable persistence, rate limiting, secret rotation, monitoring, data deletion controls, and an incident response process before handling real customer data.

05 RESULTS AND CONCLUSION

Validation, limitations, and next steps

Evaluation Results

The verified training run used 800 generated samples with seed 11 and an 80/20 stratified split. The held-out set contained 160 samples.

Metric	Value	Meaning
ROC-AUC	0.7949	Useful ranking discrimination
Accuracy	0.7375	Correct decisions at threshold 0.5
Brier score	0.1801	Probability calibration error
Automated tests	3 passed	Pipeline and API regression coverage

Limitations

The dataset is synthetic, the baseline is not causal, and subgroup fairness has not yet been evaluated. The serverless adapter should be compared against the full trained model on representative cases. The audit store is not durable in Vercel's ephemeral filesystem.

Future Work

Recommended extensions include temporal and group-aware splits, subgroup calibration, drift monitoring, durable PostgreSQL storage, human feedback capture, and an optional local Ollama or vLLM explanation writer constrained by the existing deterministic schema.

Conclusion

This project provides a complete foundation for explainable customer purchase-intention prediction. Its central contribution is not only the probability model, but the decision contract around it: intention, evidence, and reflection are returned together. This keeps the system useful for experimentation while preserving the human judgment needed for responsible customer analytics.

Project artifacts

Live workspace: https://purchase-intention-agent.vercel.app Technical report: docs/purchase_intention_project_report.pdf IEEE paper: docs/ieee_purchase_intention_paper.pdf